

Updates for week 3

In this week we mostly extended our UML diagram to adhere to the SRP, OCP and DIP and implementation methods we will use to program the project such as classes to store Material and Product and a UI among other implementation details.

Product- and Material Service

In week two the Product- and Material services were setup to have a lot of duplicate code to combat that and adhere to OCP a CREATE-RETRIEVE-UPDATE-DELETE (CRUD) generic interface is made with an abstract to generalize the CRUD code to make it easier to change the behaviour of both services.

Long term storage (repositories) and memory (registries)

A repository is added to store the products and materials in some form of Data which is TBD. Registries are used to store the products and materials for the current session like memory. Both are implemented with a generic interface to generalize it for Product and Material.

Environmental Service

To adhere to the SRP the environmentalService class is split into the EnvironmentalImpactCalculation and RecyclingGuidance classes.

The EnvironmentalImpactService is going to use different calculation strategies so to respect the OCP an interface is made so it is easy to implement new calculation strategies by just creating a new class. Furthermore we also made an interface for the recycling guidance in case there will be different guides needed in the future.

UI

The UI class is going to have other subUI it is going to use in order to keep the responsibilities separate and organized. This would include a UI for the CRUD functions for Material and Product and for recycling and environmental impact calculation.

How this improves the design

These changes should improve the structure of the project and allow for expandability and readability when implementing concrete parts. Though the overall complexity of the project increased it should be easier to implement.